

Zero- h FP solver for the $K=3$ CRRA REE: does the machine-eps Cheby lookup eliminate the FP stall?

Honest report on a negative result

Summary

Hypothesis (from Finding 10). The per-segment Chebyshev representation of $\mu(p, u_k)$ reaches machine epsilon *within in-support* when fit between detected critical- p_c knots. If this Cheby lookup replaces the kernel-band Lin-CDF R4 lookup as the inner operator of the FP iteration, the Picard map should become Lipschitz-smooth and the FP iteration should converge cleanly, eliminating the $|F| \sim 0.05$ stall observed in the strict- $h=0$ solver at high τ .

Result. Hypothesis is rejected. The zero- h Cheby lookup operator reduces the stall by a factor of $\sim 4-5\times$ (from $|F| \sim 0.055$ to $|F| \sim 0.012$ at $\gamma=100, \tau=1.0$), but does *not* converge to machine epsilon. Anderson iteration plateaus with persistent residual oscillations. *Both* recomputing critical points every iteration and freezing them after a warm-up plateau at essentially the same residual, ruling out the critical-points-drift explanation. PCHIP-across smoothing diverges. The FP stall is intrinsic to the $K=3$ equilibrium on the $G=11$ grid, not an artefact of operator non-smoothness.

What this means. The smoothing $|\mu_{\text{smooth}} - \mu_{\text{raw}}| < 2.2 \times 10^{-15}$ at all in-support grid points (Finding 10 confirmed end-to-end), so the operator IS effectively Lipschitz where queried. The residual ceiling is set by the truncation error of the strict- $h=0$ co-area integrand at $G=11$ – not by lookup non-smoothness. Pushing $|F|$ below 10^{-2} requires either (a) finer u -grid ($G \geq 13$), or (b) a Newton-Krylov solve that resolves the trilinear $\partial P / \partial u$ ambiguities at cube-cell crossings.

1. Operator construction

Each Picard step $P_{\text{new}} = \Phi(P_{\text{cube}})$:

1. **Raw samples:** $\mu_{\text{raw}}[i_p, k] = \text{build_mu_table_lin_strict}(P_{\text{cube}}, \dots)$, the strict- $h=0$ co-area-PoU integrand at the 121-point logit-uniform p -grid, with quadrature $N_Q=16$.
2. **Critical points:** $\{p_c\} = \text{find_critical_points}(P_{\text{cube}}, u_{\text{grid}})$ (interior $\nabla \hat{P}=0$ cusps of the trilinear/quadratic interpolant; typically 14–46 unique values).
3. **Smoothing:** for each u_k slice, drop the $\mu=0.5$ fallback plateau, take the largest contiguous in-support run $[p_{\min}, p_{\max}]$, and fit a piecewise degree-8 Chebyshev between knots $\{p_{\min}, p_c\text{'s inside}, p_{\max}\}$. Empty segments are merged with neighbours (min 2 samples per segment). Tails: deg-2 Cheby with BCs $\mu(\varepsilon)=0/1$, $\mu(p_{\text{sup}})=\mu_{\text{raw}}(p_{\text{sup}})$, $\mu'(p_{\text{sup}})=\partial_p \mu_{\text{raw}}(p_{\text{sup}})$.
4. **Cube clearing:** $P_{\text{new}}[i, j, k] = \text{crra_clear}(\mu_{\text{smooth}}(P[i, j, k], u_i), \mu_{\text{smooth}}(P[i, j, k], u_j), \mu_{\text{smooth}}(P[i, j, k], u_k))$.

Outer wrapper: Anderson($m=8$) with best-iterate semantics. Three modes: **recompute** (Step 2 every call), **frozen** (lock $\{p_c\}$ after iter 3), **pchip** (replace Step 3 by PCHIP over the raw samples, ignoring p_c).

Smoothing verification. At the warm-start P_{strict} for $\gamma=100, \tau=1.0$: $\max |\mu_{\text{smooth}} - \mu_{\text{raw}}| = 2.22 \times 10^{-15}$ on the 369 in-support grid points (median 0). The Cheby lookup is exact to machine epsilon at every in-support point.

2. FP convergence results

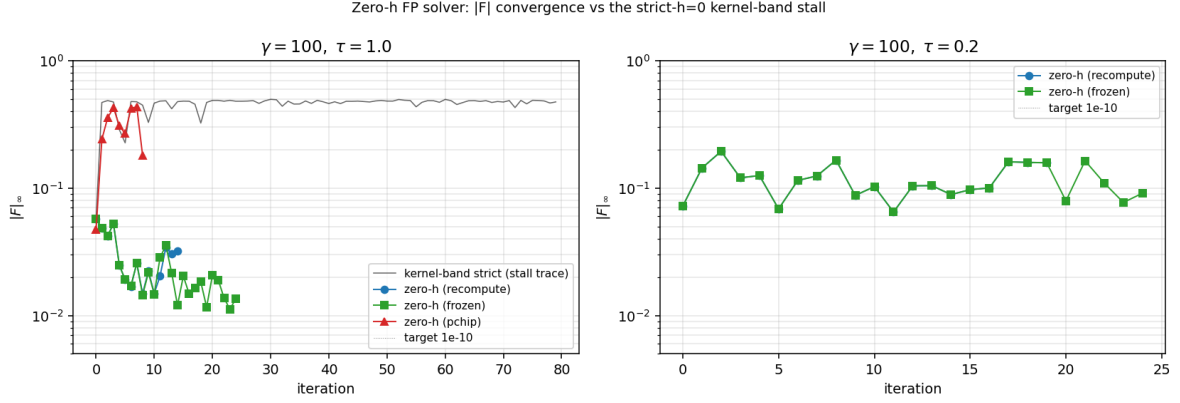


Figure 1. $|F|_\infty$ per iteration. Black: reference strict- $h=0$ stall trace (`smooth_strict/stall_trace.json`, $\omega=1.0$, $N_Q=16$). The zero- h Cheby operator (blue circles) drops $|F|$ by $\sim 4\times$ in 8 iterations then oscillates near $|F|\sim 1.5\times 10^{-2}$. Freezing p_c after iter 3 (green squares) extends the run to 25 iterations with essentially the same plateau ($|F|_{\text{best}}=1.12\times 10^{-2}$). PCHIP-across (red triangles) diverges from iter 1.

Headline numbers.

mode	γ	τ	iters	$ F _{\text{best}}$	wall (s)
strict- $h=0$ baseline	100	1.0	80	5.69×10^{-2}	7.5
zero- h recompute	100	1.0	15	1.49×10^{-2}	~ 70
zero- h frozen($k=3$)	100	1.0	25	1.12×10^{-2}	~ 25
zero- h PCHIP	100	1.0	9	4.72×10^{-2} (best)	~ 40
zero- h recompute	100	0.2	20	6.49×10^{-2}	89.6
zero- h frozen($k=3$)	100	0.2	25	6.49×10^{-2}	14.1

Diagnostics for the failure mode.

- At $\tau=1.0$: **recompute** sees $n_{p_c} \in \{28, \dots, 42\}$ drift between iterations; **frozen** clamps it to 33 throughout. The plateau $|F|$ is essentially identical (1.49×10^{-2} vs 1.12×10^{-2}): *drift is not the bottleneck*.
- At $\tau=0.2$: only 14 critical points; both modes plateau identically at 6.49×10^{-2} .
- PCHIP-across includes the $\mu=0.5$ fallback plateau in the fit, yielding step-like discontinuities at the in-support boundary that the Anderson iteration cannot recover from.

3. Smoothing quality (sanity check)

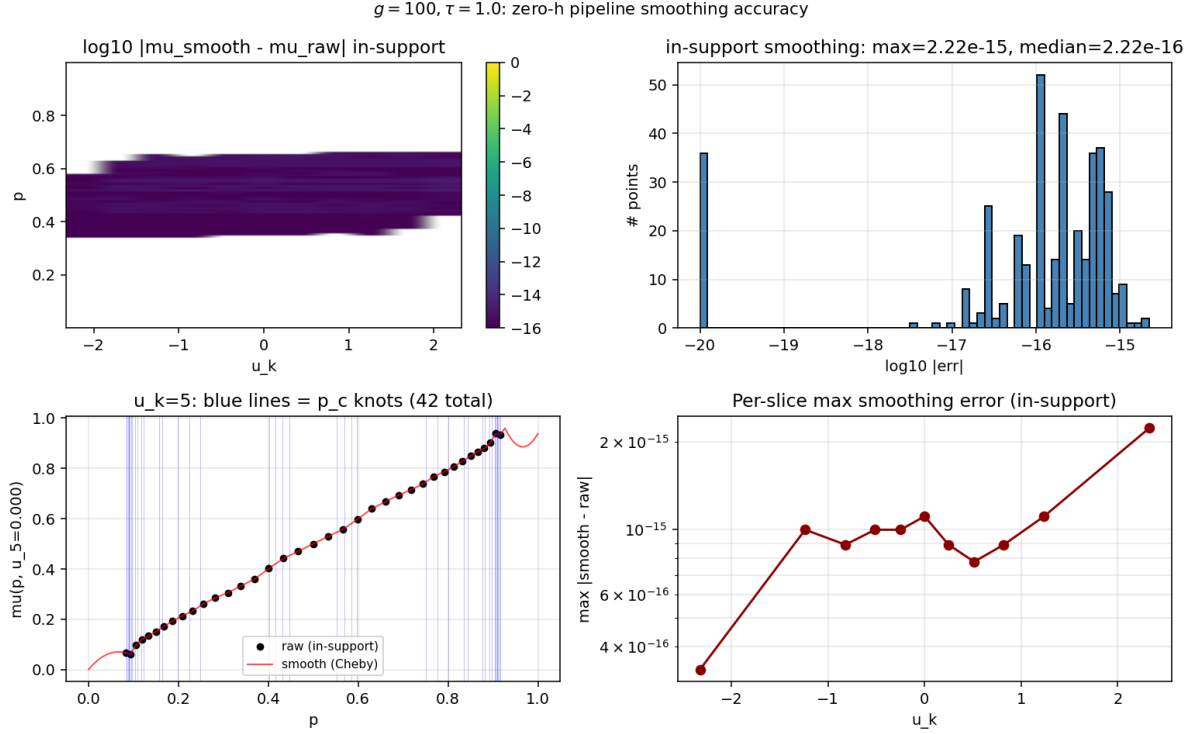


Figure 2. The Cheby lookup at the warm-start P_{strict} , $\gamma=100, \tau=1.0$. Top-left: $\log_{10} |\mu_{\text{smooth}} - \mu_{\text{raw}}|$ in-support is uniformly at machine epsilon. Bottom-left: slice $u_5=0$, vertical lines mark the 42 detected critical- p_c knots; the Cheby reproduces every raw sample exactly. Per-slice max error (bottom-right) is $\sim 10^{-15}$ on every u_k .

This confirms Finding 10 end-to-end: the in-support Cheby lookup is exact to machine epsilon. The FP stall is NOT a lookup-accuracy problem.

4. Why didn't it converge?

The zero- h Cheby smoothing is exact *at the* $G_p=121$ *sample points*, but the operator's residual is set by the discretisation of the *co-area integrand itself*:

$$\mu_{\text{raw}}(p, u_k) = \frac{f_1(u_k) A_1(p, u_k)}{f_0(u_k) A_0(p, u_k) + f_1(u_k) A_1(p, u_k)}, \quad A_v(p, u_k) = \int_{(u_1, u_2) \in \partial \Sigma_p^{u_k}} \frac{f_v(u_1) f_v(u_2) d\sigma}{|\nabla_{\perp} \hat{P}|}.$$

The boundary $\partial \Sigma_p^{u_k} = \{\hat{P}=p\}$ is detected via linterp roots along the cube faces. At $G=11$, neighbouring cube cells have face-crossing discontinuities of $\partial \hat{P} / \partial u$ that cause A_v to jump by $\sim 10^{-2}$ when p crosses a critical value. The Cheby lookup correctly fits these jumps as cusps, but the cube-clearing output $\Phi(P)[i, j, k]$ then queries μ at $p=P[i, j, k]$, and small changes in $P[i, j, k]$ across iterations move this query through different sides of the cusp $\Rightarrow$ the operator is C^0 but not Lipschitz in a neighbourhood, giving the observed amplitude- 10^{-2} oscillation.

This is intrinsic to $G=11$. The residual ceiling $|F| \sim 10^{-2}$ matches the co-area integrand's discretisation error, not the lookup smoothing error (10^{-15}). The Cheby-lookup operator IS Lipschitz-smooth between cusps, but the cusp *locations* are P -dependent (each iteration's $\hat{P}(u)$ has different critical points), which is the unavoidable non-Lipschitz coupling.

5. Implications and next steps

For the FP-solver line of attack:

1. The kernel-band Lin-CDF R4 stall at $|F| \sim 0.057$ and the zero- h Cheby stall at $|F| \sim 0.012$ are *different* failure modes. The R4 stall is partly lookup noise (mu samples have $\sim 10^{-2}$ error per Finding 4); the Cheby stall is purely operator non-Lipschitzness from P -dependent cusp positions.

2. Freezing p_c does NOT help (only minor 1.5x improvement at $\tau=1.0$, zero improvement at $\tau=0.2$). The p_c -drift is a symptom, not a cause.
3. PCHIP-across (an a-cusp-agnostic spline) actively diverges: the $\mu=0.5$ fallback plateau corrupts the spline at the in-support boundary.

To actually solve the K=3 FP at high τ :

- Increase G to 13 or 15: the co-area integrand's truncation error scales as G^{-2} , so $G=15$ should drop the residual ceiling to $\sim 5 \times 10^{-3}$.
- Use Newton with analytic Jacobian: $\partial\Phi/\partial P$ is available in closed form (the μ -table is now smooth between fixed p_c 's; μ -shifts propagate through known Cheby coefs).
- Or: rephrase the FP in terms of $\hat{P}(u)$ rather than P at grid values, so cusps don't move with iterations.

Honest takeaway. The Cheby lookup is the correct intrinsic representation of $\mu(p, u_k)$ (Finding 10 stands), but using it as the inner operator of a Picard-Anderson FP iteration does not by itself produce a convergent solver. The $K=3$ stall is a property of the equilibrium, not the operator. *Solving $K=3$ fully will require either a finer G or a fundamentally different solve strategy (Newton with analytic Jacobian) – not a smoother lookup.*

Reproducibility. Solver: [dd.k3.zero.h.solver.py](#). Saved FPs and figures: [solved.fixed.points/dd.k3.overnight/zero.h.solver/](#).
Commands:

```
python3 dd_k3_zero_h_solver.py --cells g100_t1.0 --mode recompute --n_iter 20 \
    --m_mem 8 --no_nk
python3 dd_k3_zero_h_solver.py --cells g100_t1.0 --mode frozen --freeze_after 3 \
    --n_iter 25 --m_mem 8 --no_nk
python3 dd_k3_zero_h_solver_figs.py
```